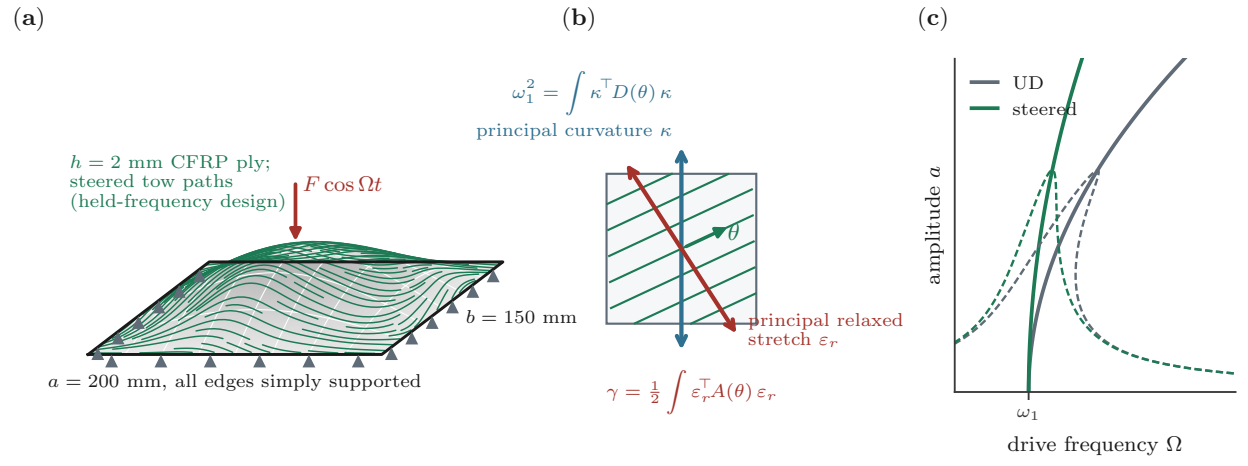

(a)
(b)
(c)
$\omega_1^2 = \int \kappa^\top D(\theta)\,\kappa$
principal curvature κ
$h = 2$ mm CFRP ply;
steered tow paths
(held-frequency design)
$F\cos\Omega t$
$b = 150$ mm
$a = 200$ mm, all edges simply supported
θ
principal relaxed
stretch ε_r
$\gamma = \frac{1}{2}\int \varepsilon_r^\top A(\theta)\,\varepsilon_r$
UD
steered
amplitude a
ω_1
drive frequency Ω